Zhi Zhang

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EDUCATION

North Carolina State University (NCSU) Jan 2024 – Present

Ph.D. Student in Computer Science

Honors: College of Engineering Summer Graduate Merit Award (GMA), 2024

New York University (NYU) Sep 2019 – Jun 2021

Master of Science in Computer Science

University of Missouri-Columbia (MU) Sep 2015 – Jun 2019

Bachelor of Science in Computer Science

Honors: Cum Laude Honors Scholar

PUBLICATIONS

Z. Zhang, C. Zhong, Y. Fu, Y. Chen, J. Jia, H. Li, C. Xu, "mV-IMU: mmWave-enabled Virtual Inertia Measurement Unit for High-fidelity Activities of Daily Living Monitoring," IEEE-EMBS International Conference on Body Sensor Networks (**BSN 2025**), **Accepted**.

Y. Fu, **Z. Zhang**, R. Ranganath, Z. Li, Y. Liu, N. Sui, H. Li, C. Xu, "TouchWave: Exploring mmWave-based Non-contact Fingertip-force Sensing in Activities of Daily Living," IEEE-EMBS International Conference on Body Sensor Networks (**BSN 2025**), **Accepted**.

R. Ranganath, **Z. Zhang**, Y. Fu, J. Jia, Y. Chen, H. Li, C. Xu, "Hand-Grip Strength Estimation through Bioacoustic Sensing," IEEE-EMBS International Conference on Body Sensor Networks (**BSN 2025**), **Accepted**.

Z. Qu, **Z. Zhang**, C. Liu, "Visual SLAM with 3D Gaussian Primitives and Depth Priors Enabling Novel View Synthesis," 2024 4th International Conference on Computer Science, Electronic Information Engineering and Intelligent Control Technology (**CEI 2024**), DOI: 10.1109/CEI63587.2024.10871531.

ACADEMIC PROJECTS

N.C. PSI Seeds Project to Improve Apple Production Jul 2025 – Present

- Contributed to mmWave radar research for precision agriculture, helping define sensing requirements for orchard fruit imaging and evaluating MIMO radar platforms and field-ready rigs.
- Designed end-to-end 2D and 3D SAR scanning workflows using planar raster and cylindrical orbital paths, developed motion-control software to execute precise scan trajectories, and collected phase-coherent datasets.
- Built the imaging pipeline: sensor & antenna/phase calibration and timing sync, motion compensation, SAR reconstruction, and registration to a common world frame for quantitative analysis.

MMW2IMU: mmWave Inertial Reconstruction Project Feb 2025 – May 2025

- Reconstructed IMU accelerations directly from mmWave radar data; device-free inertial sensing; removed pose tracking and numerical differentiation.
- Built an end-to-end pipeline: radar preprocessing, point-cloud and kinematics features, deep-learning regression with physics-guided constraints.
- Set up reproducible evaluation against wearable IMUs: synchronized recordings, fixed splits and metrics, reported strong agreement on ADL tasks.

TactileWave: Fingertip Force Dataset with mmWave Sensing Sep 2024 – Dec 2024

- Collected 13.3 hours of synchronized mmWave and ground-truth force data from 21 participants interacting with flat surfaces, daily-use items, and tools, plus ~2.9M RGB/depth frames for multimodal analysis.
- Co-designed the acquisition protocol and tools, including radar and Arduino triggering with clock synchronization, an experiment-control GUI, automated trial labeling, multi-sensor time alignment, and preprocessing for range-Doppler and point cloud features.
- Applied machine learning to analyze mmWave signals and fingertip-force relationships; engineered features and defined a reproducible evaluation protocol (data splits, metrics)

WORK EXPERIENCE

***Software Developer,* GritWorld** Dec 2022 – Dec 2023

- Developed BYD company's car HMI for dashboard and IVI (In-Vehicle Infotainment) system. Implemented dynamic animations, lane-line rendering, and data visualizations to improve driver experience.
- Developed a web server and a cloud-based service for GritGene, the company's computer graphics engine, focusing on a SaaS platform. Concentrated on backend and web server with engine asset management for 3D content (registry, versioning, permissions).
- Engineered a cloud rendering backend; automated the model-to-image/video pipeline with feature extraction and upload management, improving throughput and reliability.

***Application Developer,* ATFX Global** Feb 2022 – Oct 2022

- Contributed to the InvestingCube mobile app development, enhancing forex information aggregation with dynamic real-time updates. Implemented WebSocket and Firebase for a seamless cross-platform mobile framework and robust backend data synchronization.
- Developed a forex price tracking module with Vue for dynamic updates and strategy visualization, integrating algorithms to deliver grouped trading recommendations to users based on data analysis.
- Developed an API for a push notification service, leveraging Firebase Cloud Messaging to deliver tailored price alerts on iOS and Android devices based on user preferences.

Research Assistant, Digital Biology Lab, University of Missouri — Columbia Aug 2021 – Feb 2022

Project Link: http://www.p3db.org , https://fatplants.net

- **Plant Protein Phosphorylation Database (P3DB):** Built Angular/MySQL components for a portal analyzing phosphorylation across 45 species (about 260 in vivo proteomic studies); implemented sequence-conservation tools to infer residue importance.
- Participated in building the FatPlants web portal with analysis and visualization tools for plant fat-related genes, proteins and metabolism

Research Assistant, Stowers Institute for Medical Research Jun 2019 – Sep 2019

- **Computational Biology Team:** Developed an RNA-seq differential-expression module across 21 datasets; automated request handling and data displays to streamline lab workflows.
- Designed RESTful APIs with the Express framework, utilized async/await for data retrieval from 800+ files, and integrated with a remote R Shiny server for advanced biological data analysis.
- Delivered a responsive single-page app (Material Design) integrated with LIMS to provide detailed sample information and reduce repetitive tasks for analysts.

***Business Operations*, University of Missouri-Columbia** Jun 2016– Jan 2018

Project Manager in Web Application Project in Drupal 8

- Led 7 members to design and built 8 websites for 12 university departments including Human Resource Services, Police Department, etc., achieving 2000+ daily visits shown through Google Analytics and reducing the workload of MU Operation's IT department by 30% within 1 year

Java Developer in Java Web Application Project

- Developed and maintained four web services for contract, salary, printing/mailing, and temp staff management using Vaadin7 and the MVP pattern.
- Designed database schemas, optimized data insertion and system speed using stored SQL procedures, and enhanced security with LDAP authentication.

SKILLS

Programming Languages: JavaScript, Java, Python, PHP

Back-end Frameworks: Node.js, Java EE, Spark, Drupal 8

Front-end Frameworks: HTML/CSS, React, Redux, Vue, Bootstrap, Material-UI, EJS

Others: Linux, Nginx, Apache Spark, Unity3D, Git, Docker, Oracle, MATLAB, MySQL, MongoDB